BIOLOGY

Time: 3 Hours **Total Marks: 70**

Note:

i. All questions are compulsory.
ii. Answers to Section-I and Section-II should be written in **Two Separate** answer books.
iii. Questions from Section-I attempted in the answer book of Section-II and vice-versa will not be assessed / not be given any credit.
iv. Draw neat and labelled diagrams wherever necessary.
v. Figures to the right indicate full marks.
vi. Answer to every new question must begin on a new page.

SECTION – I

[BOTANY]

Q.1. Select and write the most appropriate answer from the given alternatives for each sub-question: **[7]**

i. How many meiotic divisions will be needed to produce 44 female gametophytes in angiosperms?
(A) 11 (B) 22
(C) 44 (D) 66

ii. If the codon on m-RNA is AUG, the compatible anticodon on t-RNA is _____.
(A) UAG (B) UAC
(C) GUA (D) AUC

iii. _____ enzymes are used as biological scissors in r-DNA technology.
(A) Restriction endonucleases (B) DNA ligases
(C) DNA polymerases (D) Reverse transcriptases

iv. Vegetative propagation takes place with the help of leaves in _____ plant.
(A) *Kalanchoe* (B) *Oxalis*
(C) *Cynodon* (D) *Dahlia*

v. *Trichoderma konigi* is a source of _____ enzyme.
(A) invertase (B) lipase
(C) pectinase (D) cellulase

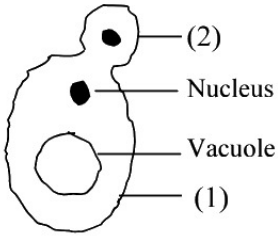
vi. What is the correct sequence of the stages in bacteriophage lytic cycle?
(A) Attachment, Penetration, Lysis, Multiplication
(B) Attachment, Penetration, Multiplication, Lysis
(C) Lysis, Penetration, Multiplication, Attachment
(D) Attachment, Lysis, Multiplication, Penetration

vii. Edible fruiting bodies are produced by _____.
(A) *Yeast* (B) *Rhizopus*
(C) *Nostoc* (D) *Agaricus*

Q.2. (A) Answer each question in 'One' sentence only: **(6)[12]**

i. Give reason : Emasculation is done in a flower which is selected as female parent.

ii. Label the parts (1) and (2) in budding of yeast cell.



iii. Enlist types of DNA library.

iv. Define parthenocarpy.

v. What is meant by ecological succession?

vi. Give the role of VAM related to soil fertility.

(B) Give diagrammatic representation of pyramid of numbers. **(2)**

(C) Attempt any TWO of the following: **(4)**

i. With the help of a suitable example, explain protandry.

ii. How depletion of ozone layer affects living organisms?

iii. Give graphic representation of cyclic photophosphorylation.

iv. Explain the role of microbes in sewage treatment.

Q.3. (A) Attempt any TWO of the following : **(6)[9]**

i. Describe the steps of PCR technique.

ii. Write a note on multiple alleles considering example of human blood groups.

iii. What is 'biofortification'? Which steps would you follow to develop a new variety of crop plant by selective breeding?

(B) Sketch and label the T.S. of undehisced anther. **(3)**

Q.4. Describe the process of glycolysis with the help of schematic representation. **[7]**

OR

Explain semi-conservative replication of eukaryotic DNA.

SECTION – II
[ZOOLOGY]

Q.5. Select and write the most appropriate answer from the given alternatives for each sub-question: **[7]**

i. Elephantiasis is caused by _____.

(A) *Wuchereria bancrofti* (B) *Plasmodium vivax*

(C) Bedbug (D) Elephant

ii. Genetically engineered human insulin is obtained by inserting the gene in _____.

(A) Pancreatic cells (B) *E. coli*

(C) *Agrobacterium tumefaciens* (D) *Drosophila melanogaster*

iii. Urea splitting bacteria are responsible for _____ types of stones.

(A) calcium oxalate (B) calcium phosphate

(C) calcium carbonate (D) struvite

iv. The connecting link between 'ape' and 'man' is _____.

(A) *Dryopithecus* (B) *Australopithecus*

(C) *Homo erectus* (D) *Homo neanderthalensis*

v. Crippling disease is caused due to poisoning of _____ .
(A) Arsenic (B) Lead
(C) Mercury (D) Selenium

vi. The genetic marker used as key factor in DNA fingerprinting is _____.
(A) VNTRs (B) Exons
(C) Introns (D) DNA probes

vii. The trophoblast cells in contact with embryonal knob are called _____ .
(A) Inner mass cells (B) Blastomere
(C) Amniogenic cells (D) Cells of Rauber

Q.6. (A) Answer the following in 'One' sentence each: **(6)[12]**

i. Mention any 'two' methods used to prevent spoilage of fish.
ii. Define the term 'genome'.
iii. What is meant by organic evolution?
iv. What is criss-cross inheritance?
v. Mention any 'two' functions of uterus.
vi. What is Red Data Book?

(B) Sketch and label the structure of chromosome. **(2)**

(C) Attempt any TWO of the following: **(4)**

i. Explain the concept 'survival of the fittest'.
ii. Give economic importance of apiculture.
iii. Give the importance of transgenic animals.
iv. Explain the mechanism of sex determination in human being.

Q.7. (A) Attempt any TWO of the following: **(6)[9]**

i. Describe T.S. of the spinal cord.
ii. Explain the structure of antibody.
iii. Give the effects of air pollutants on human health.

(B) Sketch and label L.S. of human kidney. **(3)**

Q.8. With the help of a labelled diagram describe the internal structure of human heart. **[7]**

OR

Describe the stages during three trimesters of pregnancy. Explain the process of parturition. Add a note on lactation.